

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

MECHANICAL PROPERTIES OF SOLIDS

LECTURE-02 **PHYSICS**

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Topics To Be Covered

- 1 Revision dec. no. ①
- ✓ 2 Strain
- ✓ 3 Types of Strain
- ✓ 4 Hookes Law
- 5 Elastic Potential Energy

Recap From 1st Lecture

Properties of Solid

- Maximum IAF 
- minimum IAD 

External Force

Restoring Force

Elasticity

Steel >>> Rubber

Property to regain its shape

Plasticity

Do not regain its shape

Stress

- The Restoring force acting per unit Area

$$\text{Stress} = \frac{F_R}{A} = \frac{N}{m^2} \text{ (Pascal)}$$

⇒ Longitudinal Stress / Normal Stress

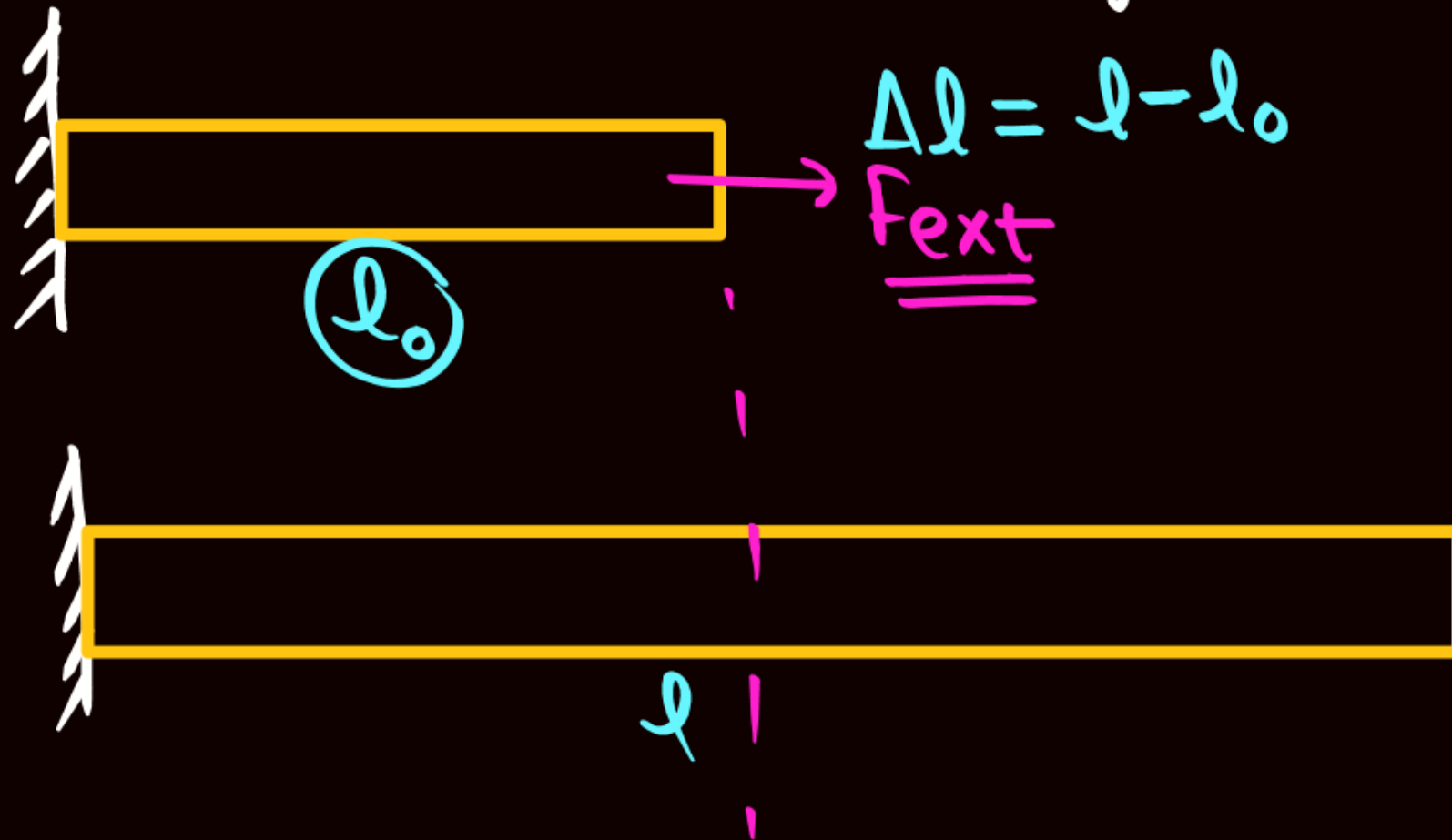
⇒ Volumetric Stress / Bulk Stress

⇒ Tangential Stress / Shear Stress

$$\text{STRAIN} = \frac{\Delta l}{l}$$

lecture no. 2

Strain:- External Force $\rightarrow \div$ Δl
It is the Ratio of change in Configuration
to the Original Configuration (l_0)



$$\text{Strain} = \frac{\Delta l}{l_0}$$

$$\# \text{ Strain} = \frac{\Delta l}{l_0} = \frac{\cancel{\text{m}}}{\cancel{\text{m}}} = 1 \text{ (unitless)}$$

$$\# \text{ Strain} = \frac{\Delta l}{l_0} = \frac{K}{K} = 1 \text{ (Dimensionless)}$$

Strain

Imp

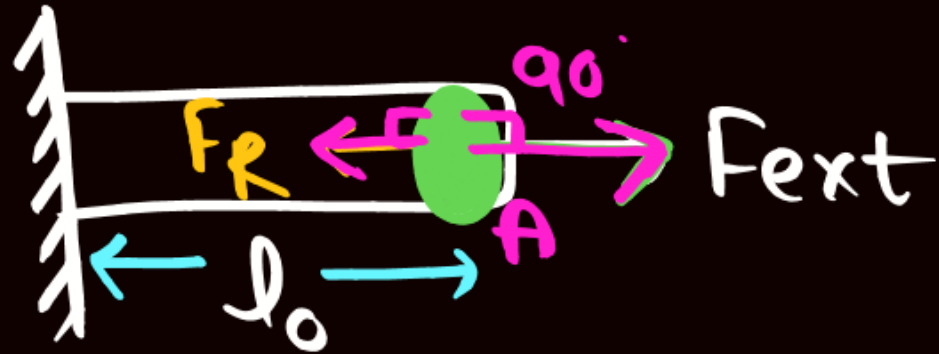
① longitudinal Strain
OR
Normal Strain

② Volumetric Strain
OR
Bulk Strain

③ Shear Strain
OR
Tangential Strain

① Longitudinal Strain / Normal Strain

- The Strain caused due to longitudinal stress



$$\boxed{\Delta l = l - l_0}$$

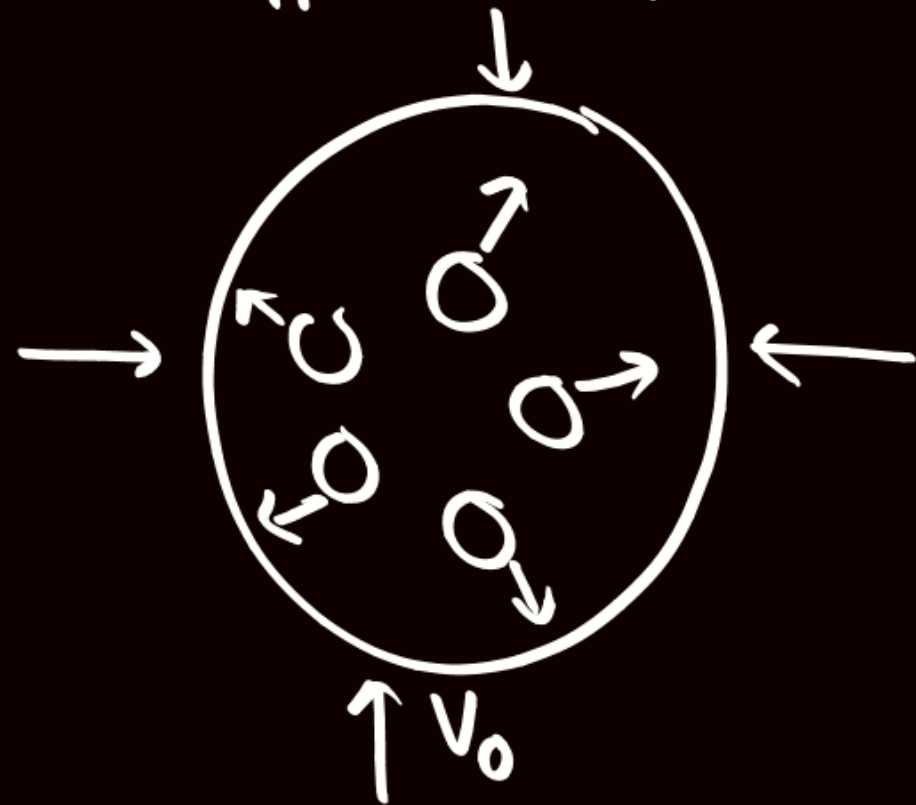
$$LS = \frac{\Delta l}{l_0}$$

$$LS = \frac{l - l_0}{l_0}$$



② Volumetric Strain / Bulk Strain

The Strain Caused by Volumetric Stress



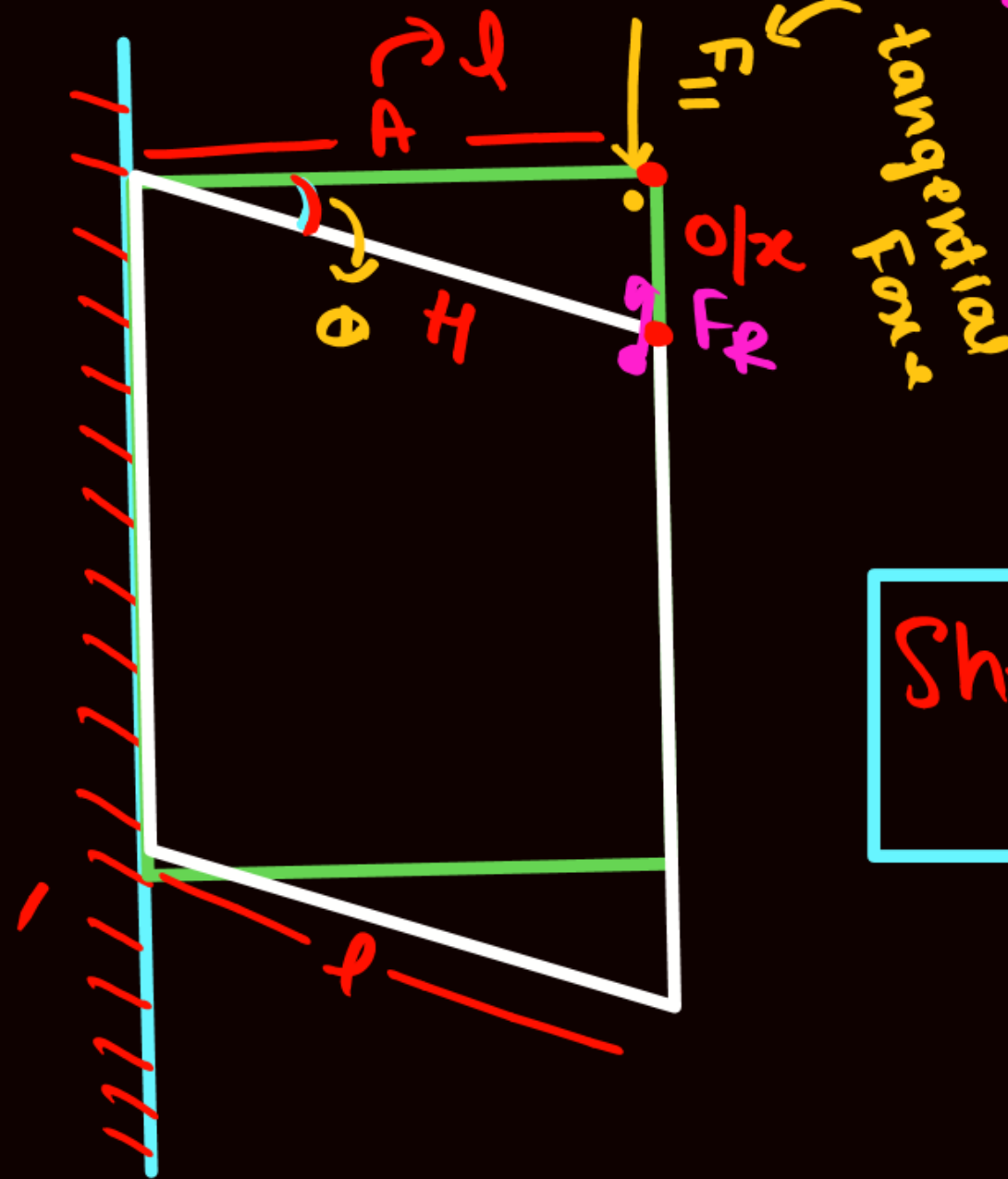
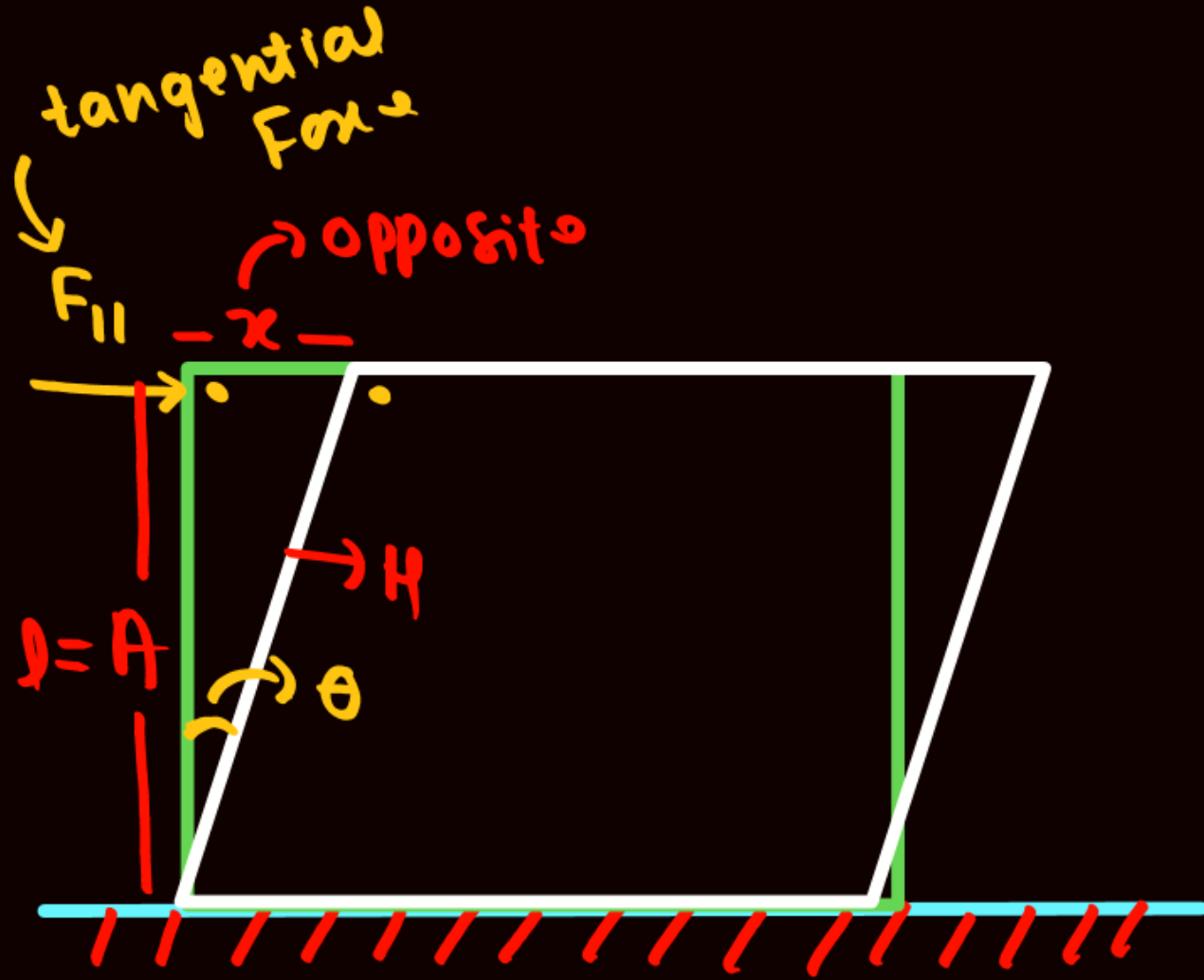
$$\Delta V = V - V_0$$

$$VS = \frac{\Delta V}{V_0} = \frac{V - V_0}{V_0}$$

③ Shear Strain / tangential Strain (jee-2023)

This Strain develop due to Shear Stress

Shear Strain = $\tan \theta$



Shear Strain = $\frac{x}{A}$

Shear Strain = $\frac{x}{A}$

Shear Strain = $\frac{x}{\phi}$

(Hooke's law)

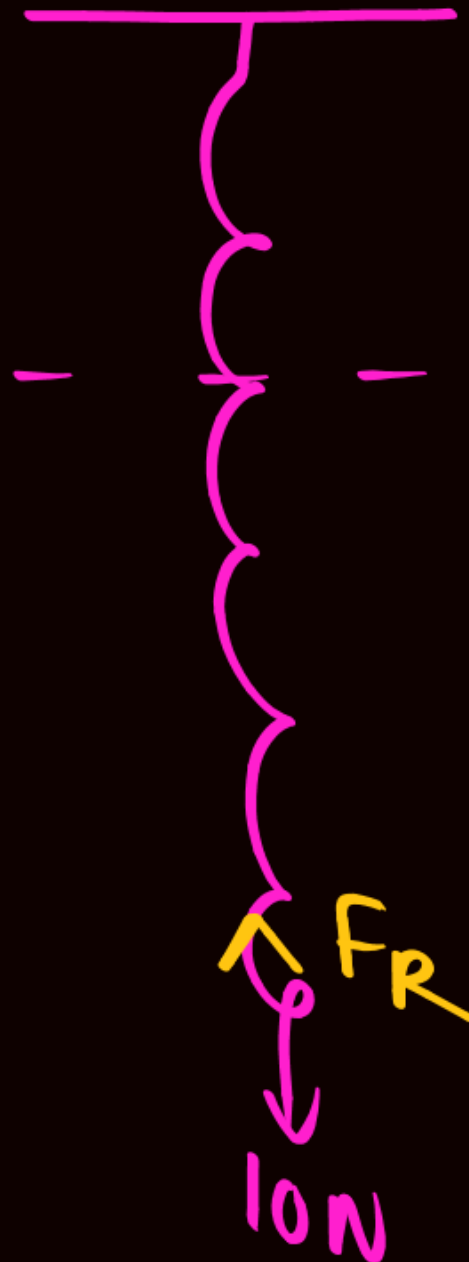
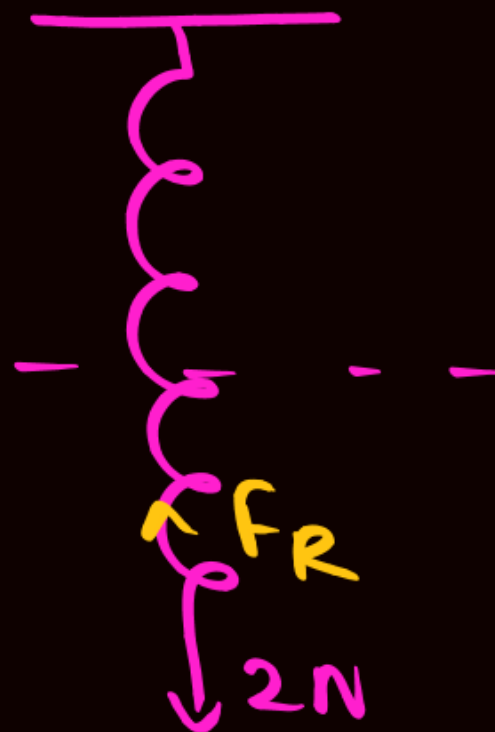
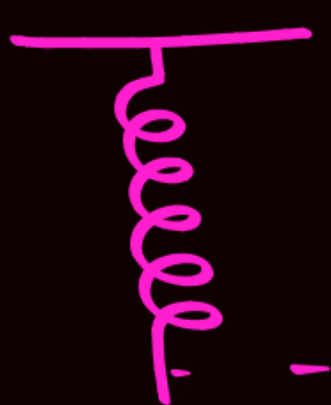
- It gives the Relationship b/w Stress and Strain

Stress $\propto$ Strain $\Rightarrow$ external force

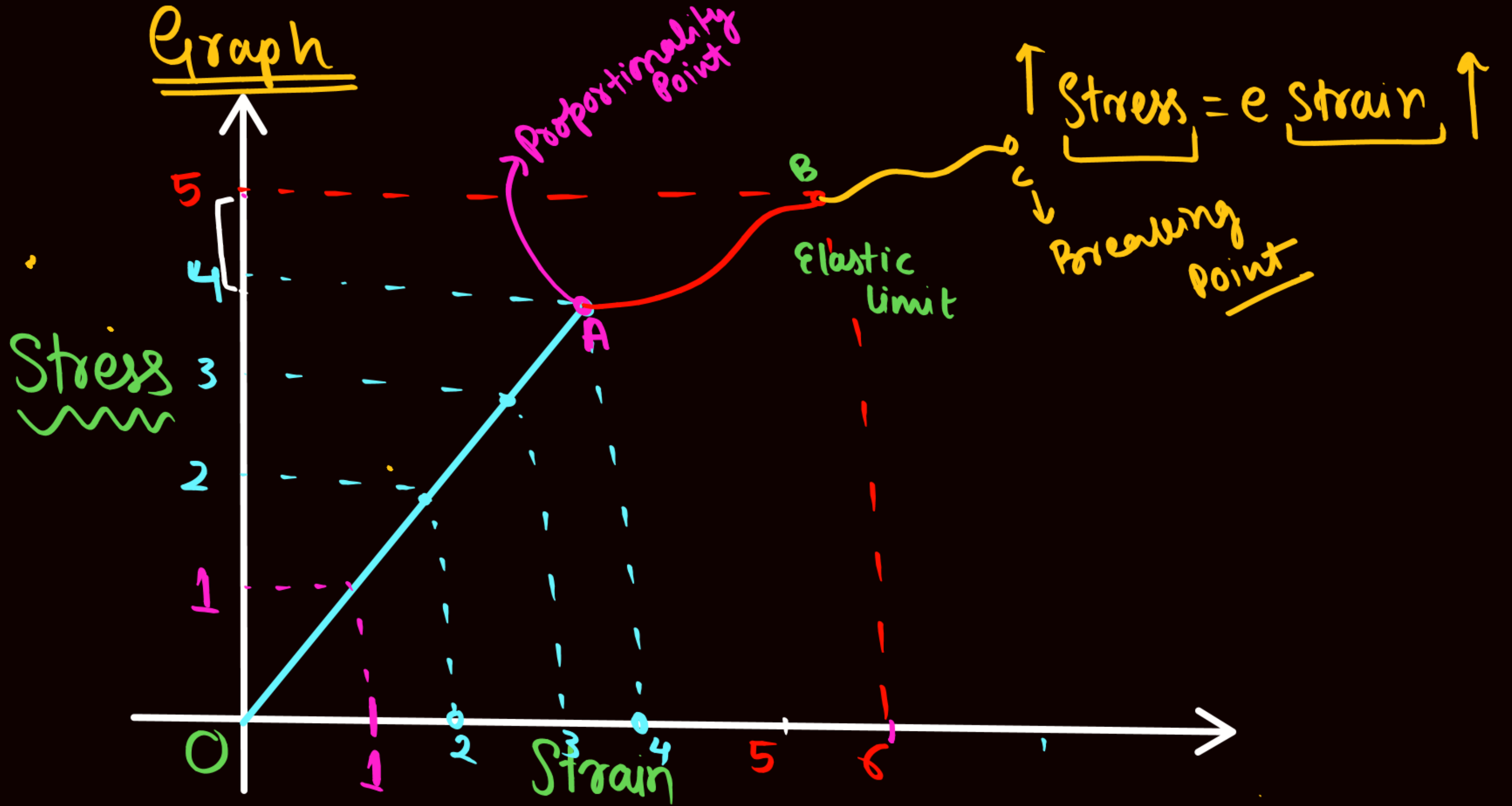
$$\text{Stress} = e \text{ Strain}$$



modulus of elasticity



Graph



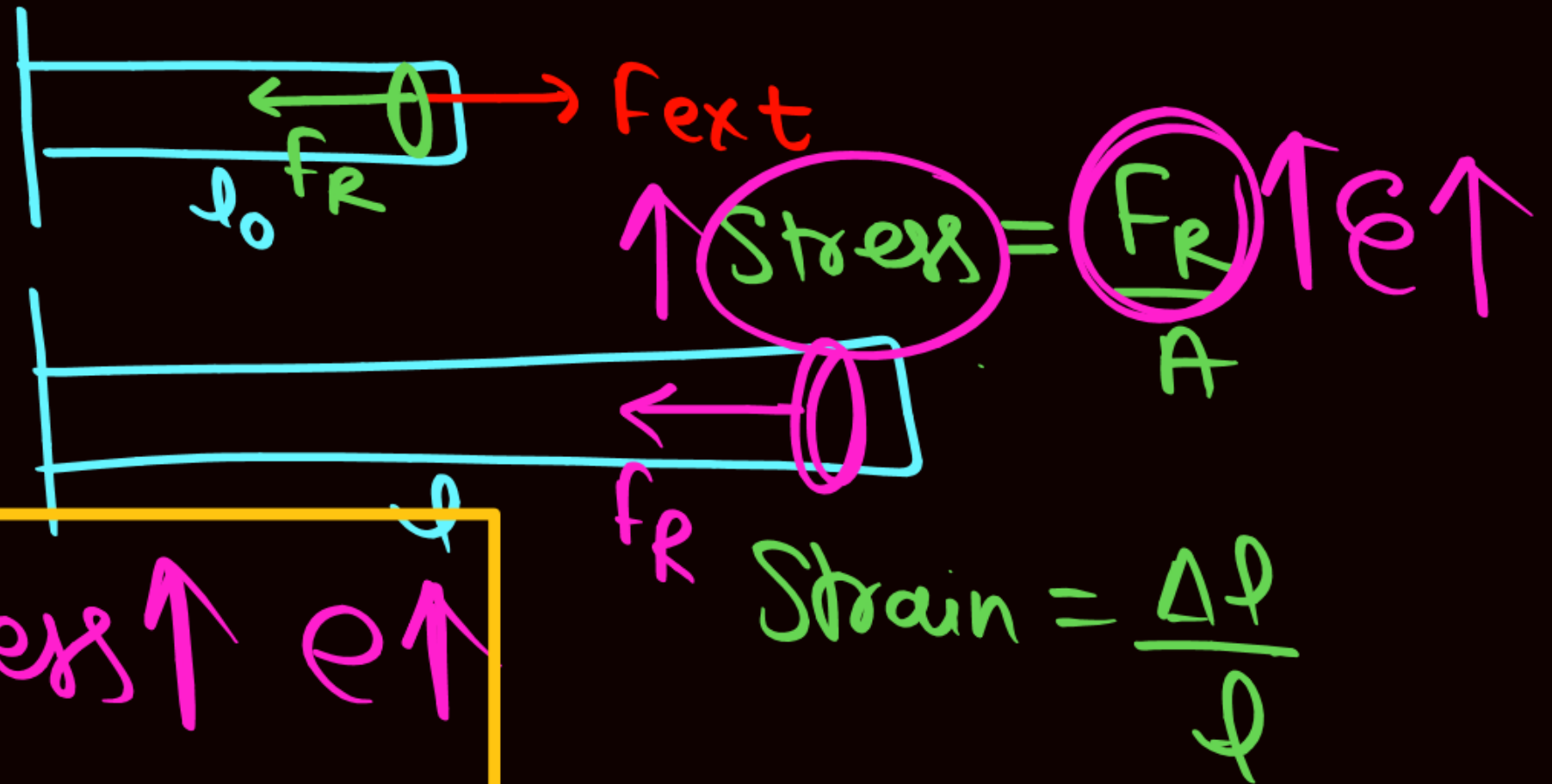
Modulus of elasticity (e)

It defines elasticity of material

modulus of elasticity is a Ratio of Stress to Strain

Elasticity $\uparrow\uparrow$

$$e = \frac{\text{Stress}}{\text{Strain}}$$



Elasticity $\uparrow$ $F_R \uparrow$ Stress $\uparrow$ $e \uparrow$

$\rho = \text{Stress}$

$\Downarrow$

Strain = 3

3 Types

Modulus Of Elasticity (e) ✓

★ Imp

$$1. \text{ Young modulus of } e (\gamma) = \frac{\text{✓ Longitudinal Stress}}{\text{Longitudinal Strain}}$$

$$2. \text{ Bulk modulus of } e (\beta) = \frac{\text{Volumetric Stress}}{\text{V Strain}}$$

$$3. \text{ Shear modulus of elasticity } = \frac{\text{Shear Stress}}{\text{Shear Strain}}$$

modulus of Rigidity (η)

Young modulus of elasticity (Y)

It is the Ratio of longitudinal stress to the longitudinal strain

$$Y = \frac{LS_{\text{stress}} \frac{a}{b}}{LS_{\text{strain}} \frac{c}{d}} = \frac{a \times d}{b \times c}$$

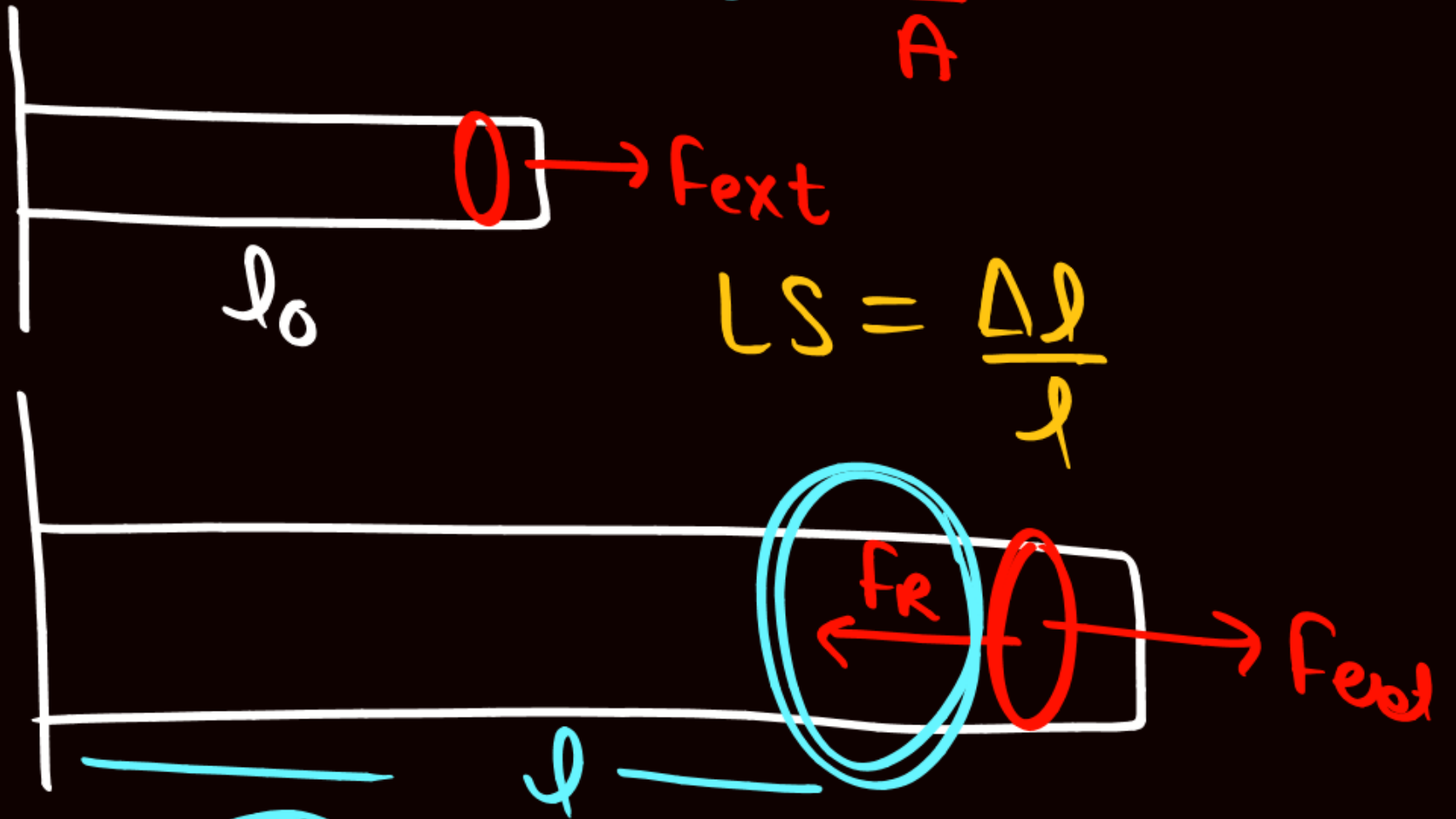
$$Y = \frac{\frac{F}{A}}{\frac{\Delta l}{l}}$$

$$Y = \frac{F}{A} \times \frac{l}{\Delta l}$$

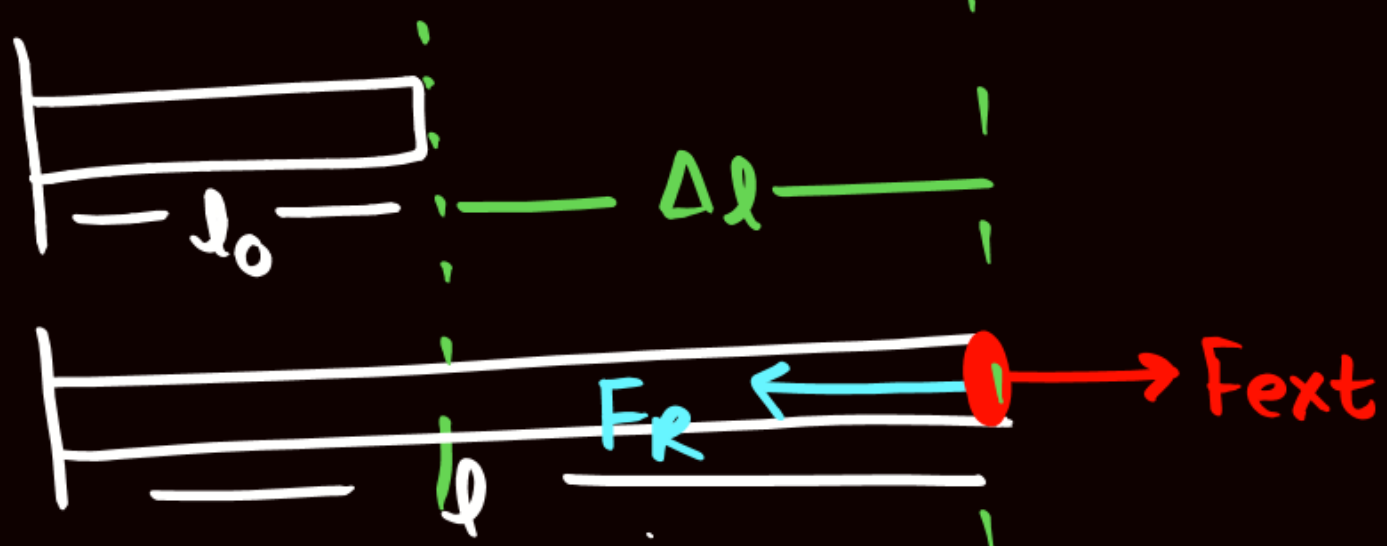
later use

$$LS = \frac{F_R}{A}$$

$$LS = \frac{\Delta l}{l}$$



Restoring force in Rod (Neet) 2023



$l = l_0 + \Delta l$

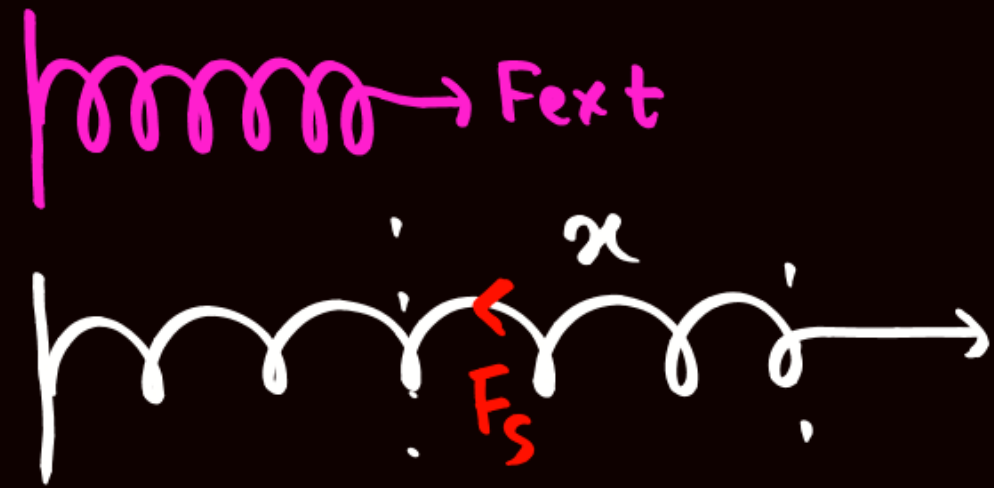
$l - l_0 = \Delta l$

$F_R \propto \Delta l$

$F_R = -K \Delta l$

Rod Constant

Restoring force in Spring



$F_S \propto x$

$F_S = -Kx$

Spring force

or
Restoring force in Spring

Spring Constant

Q // Find out the value of Rod Constant (K) ✓ CBSE

Sol $F_R = \underline{K} \Delta l$ — ①

$$Y = \frac{F_R}{A} \times \frac{l}{\Delta l}$$

$$Y = \frac{L \text{ Stress}}{L \text{ Strain}}$$

$$Y = \frac{\frac{F_R}{A}}{\frac{\Delta l}{l}}$$

$$Y = \frac{F_R}{A} \times \frac{l}{\Delta l}$$

$$Y A \Delta l = F_R l$$

$$\frac{Y A}{l} \Delta l = F_R$$

$$F_R = \frac{Y A}{l} \Delta l$$

$$F_R = \underline{(K)} \Delta l$$

#

$$K = \frac{Y A}{l}$$

Poisson Ratio

It is the negative of ratio of lateral strain to longitudinal stress
Shear Strain

Elastic Potential Energy

The potential energy store when you stretch the rod.

HOMEWORK

Neet
2021

on increasing temperature (e)

modulus
of elasticity

(a) increase

(b) decrease

(c) same

(d) main gacda hu

*"The secret to getting ahead is
getting started."*



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